

LINEAR REGRESSION LAB 1

SHUBHAM KUMAR

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```
weight_1<-c(165,167,180,155,212,175,190,210,200,149,158,169,170)
systolic_BP1<-c(130,133,150,128,151,146,150,140,148,125,133,135,150)
weight_2<-c(172,159,168,174,183,215,195,180,143,240,235,192,187)
systolic_BP2<-c(153,128,132,149,158,150,163,156,124,170,165,160,159)
```

#manually for weight 1 and systolic bp 1

#calculate B1 slope

```
mean_weight1<-mean(weight_1)
mean_weight1
```

```
## [1] 176.9231
```

```
mean_bp<-mean(systolic_BP1)
mean_bp
```

```
## [1] 139.9231
```

```
sum_w1<-sum((weight_1-mean_weight1)*(systolic_BP1 - mean_bp))
sum_w1
```

```
## [1] 1693.923
```

```
beta1<-sum_w1/sum((weight_1 - mean_weight1)^2)
beta1
```

```
## [1] 0.3380461
```

#calculate Bo intercepts

```
beta_note<-mean_bp - beta1*mean_weight1
beta_note
```

```
## [1] 80.11492
```

#check

```
lm(systolic_BP1 ~ weight_1)
```

```
##
```

```
## Call:
```

```
## lm(formula = systolic_BP1 ~ weight_1)
```

```
##
```

```
## Coefficients:
```

```
## (Intercept)      weight_1
```

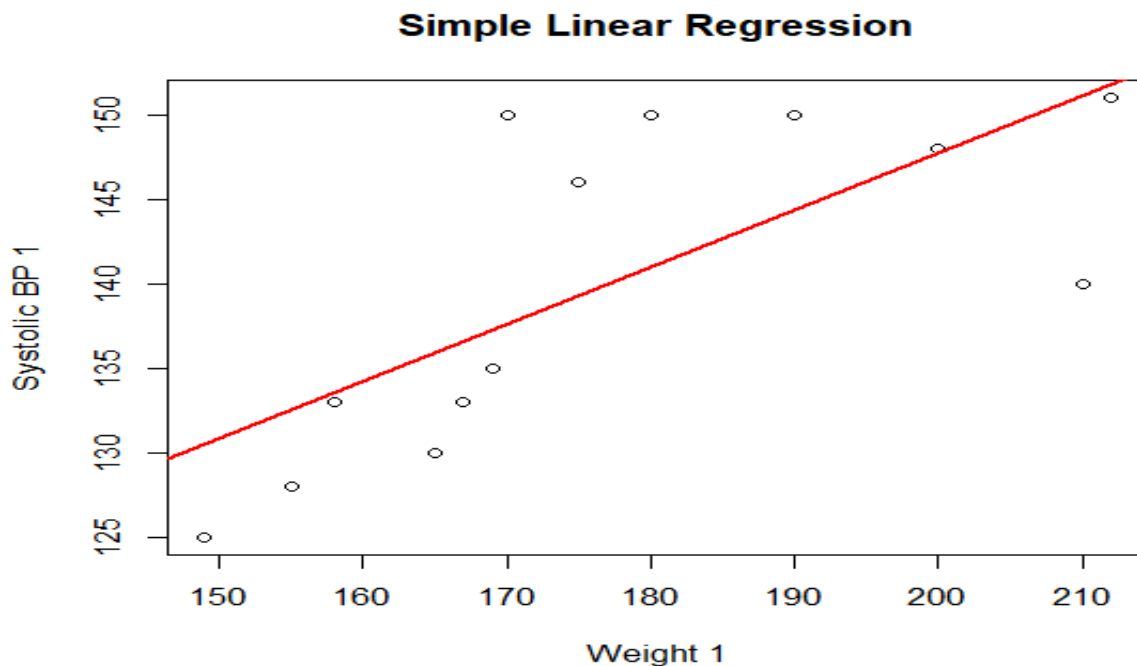
```
##      80.115         0.338
```

#plot the scatter plot for with regression line

```
simplelinear1<-lm(systolic_BP1 ~ weight_1)
```

```
plot(systolic_BP1 ~ weight_1  
     main = "Simple Linear Regression",  
     xlab = "Weight_1",  
     ylab = "Systolic BP_1")
```

```
abline(simplelinear1, col = "red", lwd = 2)
```



#manually for weight 2 and systolic bp 2

#calculate B2 slope

```
mean_weight2<-mean(weight_2)  
mean_weight2
```

```
## [1] 187.9231
```

```
mean_bp2<-mean(systolic_BP2)  
mean_bp2
```

```
## [1] 151.3077
```

```
sum_w2<-sum((weight_2-mean_weight2)*(systolic_BP2 - mean_bp2))  
sum_w2
```

```
## [1] 3914.308
```

```
beta2<-sum_w2/sum((weight_2 - mean_weight2)^2)  
beta2
```

```
## [1] 0.4113862
```

```
#calculate Bo intercepts
```

```
beta_note2<-mean_bp2 - beta2*mean_weight2  
beta_note2
```

```
## [1] 73.99874
```

```
#check
```

```
simplelinear2<-lm(systolic_BP2 ~ weight_2)  
simplelinear2
```

```
##
```

```
## Call:
```

```
## lm(formula = systolic_BP2 ~ weight_2 )
```

```
##
```

```
## Coefficients:
```

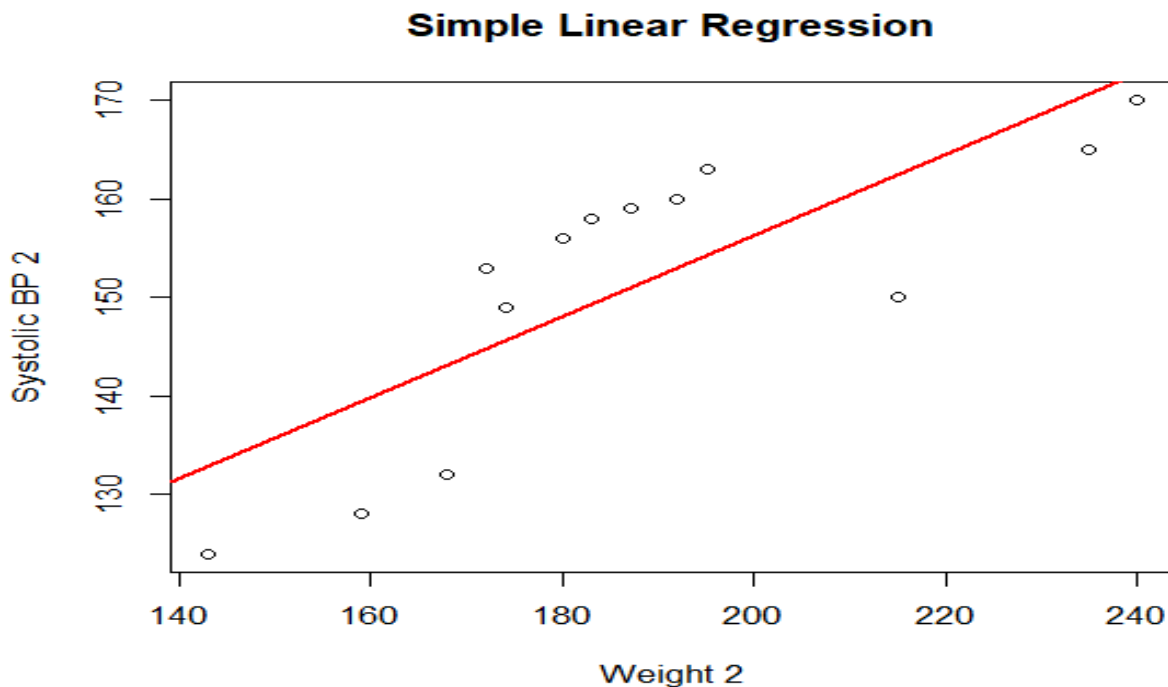
```
## (Intercept)      weight_2
```

```
##      73.9987      0.4114
```

```
#plot the scatter plot for with regression line
```

```
plot(systolic_BP2 ~ weight_2,  
     main = "Simple Linear Regression",  
     xlab = "Weight_2",  
     ylab = "Systolic BP_2")
```

```
abline(simplelinear2, col = "red", lwd = 2)
```



CONCLUSION :

Weight 1 and systolic Bp 1 :

Coefficients:

(Intercept)	weight_1
80.115	0.338

Weight 2 and Systolic Bp 2 :

Coefficients:

(Intercept)	weight_2
73.9987	0.4114

- In Both Model weight and Systolic Bp has positive correlation.
- When weight is increases Blood Pressure also increases.
- On the basis of given data we concluded that weight has strongly affects on Blood Pressure.